

Entity Types & Entity Sets

Q. Define Entity, Entity Type, and Entity Set. Clearly differentiate between a Strong Entity and a Weak Entity with examples.

📌 Definition to Remember

An **Entity** is a single real-world object that has independent existence and can be uniquely identified. An **Entity Type** is a category of entities sharing the same attributes (the blueprint), and an **Entity Set** is the actual collection of all instances of that entity type stored in the database.

1. Entity, Entity Type, and Entity Set

- **Entity:** A single, specific, real-world object that can be uniquely identified.
- *Example:* The specific person "John Doe" or a specific car with plate "ABC-123".
- **Entity Type:** A category/template for entities sharing the same attributes. Represented by a **Rectangle** in ER diagrams.
- *Example:* **STUDENT** is an entity type — all students share Name, Age, and Roll Number.
- **Entity Set:** The actual collection of all entities of a particular type in the database at a given moment.
- *Example:* All 500 currently enrolled students form the **STUDENT** entity set.

2. Strong Entity vs. Weak Entity

Entities are classified based on their ability to be uniquely identified using their own attributes.

Strong Entity

- **Definition:** An entity type that possesses its own **Primary Key** to uniquely identify each entity.
- Has an **independent existence** — does not rely on any other entity.
- Represented by a **Single Rectangle** in the ER diagram.
- **Example:** **EMPLOYEE** — uniquely identified by **Employee_ID**. Exists independently.

Weak Entity

- **Definition:** An entity type that **does not have its own Primary Key** and cannot be uniquely identified by its own attributes alone.
- **Depends** entirely on a Strong Entity (**Identifying/Owner Entity**) for its existence.
- Its key = Identifying Entity's PK + its own **Partial Key (Discriminator)**.
- Represented by a **Double Rectangle**. Its identifying relationship uses a **Double Diamond**.
- **Example:** **DEPENDENT** (employee's child). Two employees may both have a dependent named "Sarah" — Sarah can only be identified using **Employee_ID + Name**. If the Employee is deleted, the Dependent record is also deleted.

[EMPLOYEE] ==<< DEPENDENTS >>== [[DEPENDENT]]
 (Strong) Double Diamond (Weak)
 Single Rect Double Rect

3. Summary Table

Feature	Strong Entity	Weak Entity
Primary Key	Has its own Primary Key	Has only a Partial Key (Discriminator)
Existence	Independent	Depends on a Strong Entity
ER Symbol	Single Rectangle	Double Rectangle
Relationship	Single Diamond	Double Diamond (Identifying Relationship)
Example	EMPLOYEE (identified by Emp_ID)	DEPENDENT (identified by Emp_ID + Name)

★ Must-Write Points (for 10 marks)

1. Entity = single specific object; Entity Type = category template; Entity Set = all instances in DB.
2. Entity Types are represented by Rectangles in ER diagrams.
3. Strong Entity has its own Primary Key and exists independently.
4. Weak Entity has no Primary Key — uses a Partial Key + Owner's PK for identification.
5. Weak Entity is represented by a Double Rectangle; its relationship by a Double Diamond.
6. If the Strong (Owner) Entity is deleted, the associated Weak Entity must also be deleted.
7. Example: **EMPLOYEE** (strong), **DEPENDENT** (weak — relies on Employee_ID).

⚡ Quick Recall

Entity (object) → Entity Type (blueprint/Rectangle) → Entity Set (all instances) → Strong (own PK, single rect) → Weak (partial key, double rect, double diamond)